

the Discrete Fourier Transform

1 The Discrete Fourier Transform

- roots of unity
- definition of the DFT

2 Convolutions and the DFT

- convolutions become componentwise products
- applying the discrete Fourier transform

3 Interpolation by the DFT

- the DFT interpolation theorem
- applied to filter design

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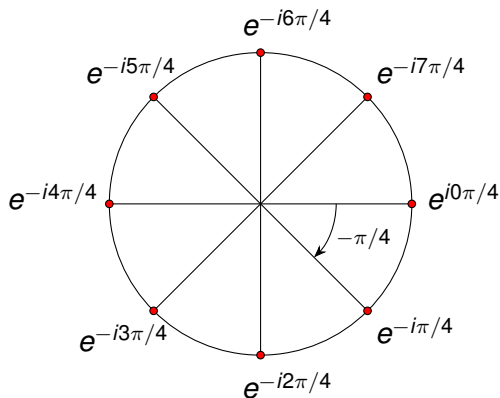
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roots of unity

The eight roots of unity, generated by $\omega = e^{-i2\pi/8} = e^{-i\pi/4}$, are



$$\omega = e^{-i\pi/4} = \cos(\pi/4) - i \sin(\pi/4), \quad i = \sqrt{-1}, \quad (\text{Euler's formula})$$

roots of unity and primitive roots

Definition (roots of unity)

The number z is an *n th root of unity* if $z^n - 1 = 0$.

Definition (primitive root of unity)

An n th root of unity is *primitive* if it is not a k th root of unity for any $k < n$.

Exercise 1:

For $n = 8$, write all n th roots that are primitive.

Verify that for each primitive root z , all other eight roots can be generated by taking powers.

sums of powers of the n th root of unity

For $n > 0$, the n th primitive root of unity is $\omega = e^{-i2\pi/n}$.

ω is a root of the equation

$$x^n - 1 = 0,$$

the other $n - 1$ roots are the powers ω^k , $k = 2, 3, \dots, n$,
with $\omega^n = \omega^0 = 1$.

The root 1 makes the polynomial $x^n - 1$ factor:

$$\omega^n - 1 = (\omega - 1) (1 + \omega + \omega^2 + \dots + \omega^{n-1}) = 0.$$

For $\omega \neq 1$, we have then

$$1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0.$$

sums of the powers of the roots of unity

As the other $n - 2$ roots are powers ω^k , $k = 1, 2, \dots, n - 1$:

$$1 + \omega^k + \omega^{2k} + \dots + \omega^{(n-1)k} = 0.$$

For $k = n$, we have

$$1 + \omega^n + \omega^{2n} + \dots + \omega^{(n-1)n} = 1 + 1 + 1 + \dots + 1 = n.$$

Proposition (the Gauss relation)

Let ω be a primitive n root of unity and let k be an integer.

$$\sum_{j=0}^{n-1} \omega^{jk} = \begin{cases} n & \text{if } k/n \text{ is an integer,} \\ 0 & \text{otherwise.} \end{cases}$$

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the discrete Fourier transform

Definition (the Discrete Fourier Transform (DFT))

Let $\mathbf{x} = [x_0, x_1, \dots, x_{n-1}]^T$ be an n -dimensional vector.

The *Discrete Fourier Transform* of $\mathbf{x}$ is $\mathbf{y} = [y_0, y_1, \dots, y_{n-1}]^T$, where

$$y_k = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j \omega^{jk}, \quad \omega = e^{-i2\pi/n}.$$

the matrix representation

$$y_k = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j \omega^{jk}$$

For example, for $n = 8$, in matrix form the DFT is

$$\begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \\ y_7 \end{bmatrix} = \frac{1}{\sqrt{n}} \begin{bmatrix} \omega^{0 \cdot 0} & \omega^{0 \cdot 1} & \omega^{0 \cdot 2} & \omega^{0 \cdot 3} & \omega^{0 \cdot 4} & \omega^{0 \cdot 5} & \omega^{0 \cdot 6} & \omega^{0 \cdot 7} \\ \omega^{1 \cdot 0} & \omega^{1 \cdot 1} & \omega^{1 \cdot 2} & \omega^{1 \cdot 3} & \omega^{1 \cdot 4} & \omega^{1 \cdot 5} & \omega^{1 \cdot 6} & \omega^{1 \cdot 7} \\ \omega^{2 \cdot 0} & \omega^{2 \cdot 1} & \omega^{2 \cdot 2} & \omega^{2 \cdot 3} & \omega^{2 \cdot 4} & \omega^{2 \cdot 5} & \omega^{2 \cdot 6} & \omega^{2 \cdot 7} \\ \omega^{3 \cdot 0} & \omega^{3 \cdot 1} & \omega^{3 \cdot 2} & \omega^{3 \cdot 3} & \omega^{3 \cdot 4} & \omega^{3 \cdot 5} & \omega^{3 \cdot 6} & \omega^{3 \cdot 7} \\ \omega^{4 \cdot 0} & \omega^{4 \cdot 1} & \omega^{4 \cdot 2} & \omega^{4 \cdot 3} & \omega^{4 \cdot 4} & \omega^{4 \cdot 5} & \omega^{4 \cdot 6} & \omega^{4 \cdot 7} \\ \omega^{5 \cdot 0} & \omega^{5 \cdot 1} & \omega^{5 \cdot 2} & \omega^{5 \cdot 3} & \omega^{5 \cdot 4} & \omega^{5 \cdot 5} & \omega^{5 \cdot 6} & \omega^{5 \cdot 7} \\ \omega^{6 \cdot 0} & \omega^{6 \cdot 1} & \omega^{6 \cdot 2} & \omega^{6 \cdot 3} & \omega^{6 \cdot 4} & \omega^{6 \cdot 5} & \omega^{6 \cdot 6} & \omega^{6 \cdot 7} \\ \omega^{7 \cdot 0} & \omega^{7 \cdot 1} & \omega^{7 \cdot 2} & \omega^{7 \cdot 3} & \omega^{7 \cdot 4} & \omega^{7 \cdot 5} & \omega^{7 \cdot 6} & \omega^{7 \cdot 7} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{bmatrix}$$

The n -by- n matrix in this representation is the *Fourier matrix*.

the Fourier matrix

For $\omega = e^{-i2\pi/n}$, the Fourier matrix is

$$F_n = \frac{1}{\sqrt{n}} \begin{bmatrix} \omega^0 & \omega^0 & \omega^0 & \dots & \omega^0 \\ \omega^0 & \omega^1 & \omega^2 & \dots & \omega^{n-1} \\ \omega^0 & \omega^2 & \omega^4 & \dots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \omega^0 & \omega^{n-1} & \omega^{2(n-1)} & \dots & \omega^{(n-1)^2} \end{bmatrix}.$$

Then for $\mathbf{x}$ and $\mathbf{y}$ two n -dimensional vectors for which

$$\mathbf{y} = F_n \mathbf{x},$$

we have that $\mathbf{y}$ is the DFT of $\mathbf{x}$.

the inverse of the Fourier matrix

$$F_n = \frac{1}{\sqrt{n}} \begin{bmatrix} \omega^0 & \omega^0 & \omega^0 & \dots & \omega^0 \\ \omega^0 & \omega^1 & \omega^2 & \dots & \omega^{n-1} \\ \omega^0 & \omega^2 & \omega^4 & \dots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \omega^0 & \omega^{n-1} & \omega^{2(n-1)} & \dots & \omega^{(n-1)^2} \end{bmatrix}$$

has as inverse (by the Gauss relation):

$$F_n^{-1} = \frac{1}{\sqrt{n}} \begin{bmatrix} \omega^0 & \omega^0 & \omega^0 & \dots & \omega^0 \\ \omega^0 & \omega^{-1} & \omega^{-2} & \dots & \omega^{-(n-1)} \\ \omega^0 & \omega^{-2} & \omega^{-4} & \dots & \omega^{-(2(n-1))} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \omega^0 & \omega^{-(n-1)} & \omega^{-2(n-1)} & \dots & \omega^{-(n-1)^2} \end{bmatrix}.$$

Observe: $F_n^{-1} = \overline{F_n}$, the complex conjugate of F .

$$\omega = e^{-i2\pi/n}$$

the inverse discrete Fourier transform

Definition (the inverse Discrete Fourier Transform (iDFT))

Let $\mathbf{y} = [y_0, y_1, \dots, y_{n-1}]^T$ be an n -dimensional vector.

The *inverse Discrete Fourier Transform* of $\mathbf{y}$ is $\mathbf{x} = [x_0, x_1, \dots, x_{n-1}]^T$,

$$\mathbf{x} = F_n^{-1} \mathbf{y},$$

where F_n is the n -by- n Fourier matrix.

Componentwise, we have the formulas:

$$x_j = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} \left(\omega^{-k} \right)^j y_k, \quad \omega = e^{-i2\pi/n}, \quad j = 0, 1, \dots, n-1.$$

verify with Julia functions

Exercise 2:

- 1 Write a Julia function `FourierMatrix` which takes on input n and which returns the Fourier matrix F_n .
- 2 Write a Julia function `inverseFourierMatrix` which takes on input n and which returns the inverse Fourier matrix F_n^{-1} .
- 3 Verify for $n = 8$ that the product of the output of your `FourierMatrix( $n$ )` with the output of your `inverseFourierMatrix( $n$ )` is indeed the identity matrix.

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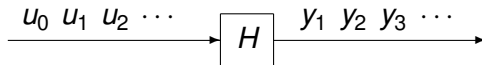
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filters and convolutions



A linear, time invariant, causal filter is determined by the impulse response $\{h_k\}_{k=0}^{\infty}$ and the transfer function is $H(z) = \sum_{k=0}^{\infty} h_k z^{-k}$.

For input u , the k -th element in the output y is

$$y_k = h_k u_0 + h_{k-1} u_1 + \cdots + h_1 u_{k-1} + h_0 u_k = \sum_{j=0}^k h_{k-j} u_j.$$

The convolution is denoted by the operator $\star$, as $y = h \star u$.

filtering a periodic signal

$$u_0, u_1, u_2, u_3, u_0, \dots \rightarrow (h_0, h_1, h_2, h_3) \rightarrow y_0, y_1, y_2, y_3, y_0, \dots$$

Let us compute $y = h \star u$, applying $y_k = \sum_{j=0}^k h_{k-j} u_j$:

$$y_0 = h_0 u_0 + h_{-1} u_1 + h_{-2} u_2 + h_{-3} u_3.$$

By the periodicity: $h_{-1} = h_3$, $h_{-2} = h_2$, and $h_{-3} = h_1$.

$$y_0 = h_0 u_0 + h_3 u_1 + h_2 u_2 + h_1 u_3$$

$$y_1 = h_1 u_0 + h_0 u_1 + h_3 u_2 + h_2 u_3$$

$$y_2 = h_2 u_0 + h_1 u_1 + h_0 u_2 + h_3 u_3$$

$$y_3 = h_3 u_0 + h_2 u_1 + h_1 u_2 + h_0 u_3$$

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applying the discrete Fourier transform

Apply $y_k = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j \omega^{jk}$, for $n = 4$, $k = 1$, on (y_0, y_1, y_2, y_3) as input:

$$\hat{y}_1 = \frac{1}{\sqrt{4}} \left(y_0 \omega^0 + y_1 \omega^1 + y_2 \omega^2 + y_3 \omega^3 \right).$$

Multiply the formulas with powers of ω :

$$\omega^0 y_0 = \omega^0 h_0 u_0 + \omega^0 h_3 u_1 + \omega^0 h_2 u_2 + \omega^0 h_1 u_3$$

$$\omega^1 y_1 = \omega^1 h_1 u_0 + \omega^1 h_0 u_1 + \omega^1 h_3 u_2 + \omega^1 h_2 u_3$$

$$\omega^2 y_2 = \omega^2 h_2 u_0 + \omega^2 h_1 u_1 + \omega^2 h_0 u_2 + \omega^2 h_3 u_3$$

$$\omega^3 y_3 = \omega^3 h_3 u_0 + \omega^3 h_2 u_1 + \omega^3 h_1 u_2 + \omega^3 h_0 u_3$$

Adding up the above left hand sides leads to

$$\omega^0 y_0 + \omega^1 y_1 + \omega^2 y_2 + \omega^3 y_3 = 2\hat{y}_1.$$

collecting terms

$$\omega^0 y_0 = \omega^0 h_0 u_0 + \omega^0 h_3 u_1 + \omega^0 h_2 u_2 + \omega^0 h_1 u_3$$

$$\omega^1 y_1 = \omega^1 h_1 u_0 + \omega^1 h_0 u_1 + \omega^1 h_3 u_2 + \omega^1 h_2 u_3$$

$$\omega^2 y_2 = \omega^2 h_2 u_0 + \omega^2 h_1 u_1 + \omega^2 h_0 u_2 + \omega^2 h_3 u_3$$

$$\omega^3 y_3 = \omega^3 h_3 u_0 + \omega^3 h_2 u_1 + \omega^3 h_1 u_2 + \omega^3 h_0 u_3$$

Adding up the above right hand sides and collecting terms gives

$$\begin{aligned} & u_0 \left(\omega^0 h_0 + \omega^1 h_1 + \omega^2 h_2 + \omega^3 h_3 \right) \\ & + u_1 \left(\omega^0 h_3 + \omega^1 h_0 + \omega^2 h_1 + \omega^3 h_2 \right) \\ & + u_2 \left(\omega^0 h_2 + \omega^1 h_3 + \omega^2 h_0 + \omega^3 h_1 \right) \\ & + u_3 \left(\omega^0 h_1 + \omega^1 h_2 + \omega^2 h_3 + \omega^3 h_0 \right) \end{aligned}$$

the Fourier transform of h_0, h_1, h_2, h_3

Apply $y_k = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j \omega^{jk}$, for $n = 4$, $k = 1$, on (h_0, h_1, h_2, h_3) as input:

$$\hat{h}_1 = \frac{1}{\sqrt{4}} \left(h_0 \omega^0 + h_1 \omega^1 + h_2 \omega^2 + h_3 \omega^3 \right).$$

Now we can rewrite the added right hand sides:

$$\begin{aligned} u_0 \left(\omega^0 h_0 + \omega^1 h_1 + \omega^2 h_2 + \omega^3 h_3 \right) &= u_0 \hat{h}_1 \\ + u_1 \left(\omega^0 h_3 + \omega^1 h_0 + \omega^2 h_1 + \omega^3 h_2 \right) &= u_1 \hat{h}_1 \omega^1 \\ + u_2 \left(\omega^0 h_2 + \omega^1 h_3 + \omega^2 h_0 + \omega^3 h_1 \right) &= u_2 \hat{h}_1 \omega^2 \\ + u_3 \left(\omega^0 h_1 + \omega^1 h_2 + \omega^2 h_3 + \omega^3 h_0 \right) &= u_3 \hat{h}_1 \omega^3 \end{aligned}$$

the Fourier transform of u_0, u_1, u_2, u_3

Apply $y_k = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j \omega^{jk}$, for $n = 4$, $k = 1$, on (u_0, u_1, u_2, u_3) as input:

$$\hat{u}_1 = \frac{1}{\sqrt{4}} \left(u_0 \omega^0 + u_1 \omega^1 + u_2 \omega^2 + u_3 \omega^3 \right).$$

So we found

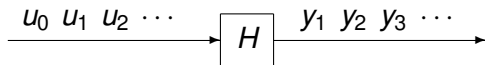
$$\begin{aligned} 2\hat{y}_1 &= u_0 \hat{h}_1 + u_1 \hat{h}_1 \omega^1 + u_2 \hat{h}_1 \omega^2 + u_3 \hat{h}_1 \omega^3 \\ &= \hat{h}_1 \left(u_0 + u_1 \omega^1 + u_2 \omega^2 + u_3 \omega^3 \right) \\ &= \hat{h}_1 2\hat{u}_1, \end{aligned}$$

or, more in general:

$$\hat{y}_1 = \sqrt{n} \hat{h}_1 \hat{u}_1,$$

where $\hat{y} = \text{DFT}(y)$, $\hat{h} = \text{DFT}(h)$, and $\hat{u} = \text{DFT}(u)$.

the DFT convolution property



The filter H has impulse response $\{h_k\}_{k=0}^{\infty}$.

$$\begin{array}{ccc} y = h \star u & \xrightarrow{\text{DFT}} & \hat{y} = \sqrt{n} \hat{h} \cdot \hat{u} \\ \text{convolution} & & \text{componentwise product} \end{array}$$

where $\hat{y} = \text{DFT}(y)$, $\hat{h} = \text{DFT}(h)$, and $\hat{u} = \text{DFT}(u)$.

Theorem (the DFT convolution property)

The discrete Fourier transform of $h \star u$ is $\sqrt{n}$ times the componentwise product of the discrete Fourier transforms of h and u .

verify numerically and symbolically

Exercise 3:

Verify the DFT convolution property on two random vectors $\mathbf{x}$ and $\mathbf{y}$, for $n = 8$.

- 1 Use your `FourierMatrix` of Exercise 2 to compute the DFT of $\mathbf{x}$ and $\mathbf{y}$, $\hat{\mathbf{x}} = \text{DFT}(\mathbf{x})$ and $\hat{\mathbf{y}} = \text{DFT}(\mathbf{y})$.
- 2 Verify that $\sqrt{8}$ times the componentwise product of $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ equals the DFT of $\mathbf{x} \star \mathbf{y}$.

Exercise 4:

We derived the statement of the DFT convolution property for $n = 4$ and $k = 1$.

Verify the DFT convolution property by symbolic calculation for $n = 4$ and $k = 2$.

the DFT convolution theorem

Theorem (the DFT convolution property)

Let $\mathbf{x}$ and $\mathbf{y}$ be two n -dimensional vectors.

The discrete Fourier transform of $\mathbf{x} \star \mathbf{y}$ is $\sqrt{n}$ times the componentwise product of the discrete Fourier transforms of $\mathbf{x}$ and $\mathbf{y}$.

Theorem (the DFT convolution theorem)

Let $\mathbf{x}$ and $\mathbf{y}$ be two n -dimensional vectors.

The convolution $\mathbf{x} \star \mathbf{y}$ can be computed as

$$\mathbf{x} \star \mathbf{y} = \text{iDFT}(\sqrt{n} \text{DFT}(\mathbf{x}) \cdot \text{DFT}(\mathbf{y})),$$

where DFT is the discrete Fourier transform and

iDFT is the inverse discrete Fourier transform.

$\cdot$ is the componentwise product of two vectors.

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the DFT interpolation theorem

Theorem (the DFT Interpolation Theorem)

Consider n points $t_j = j/n$, for $j = 0, 1, \dots, n-1$.

Let $\mathbf{x} = [x_0, x_1, \dots, x_{n-1}]^T$, $\mathbf{y} = F_n \mathbf{x}$, where F_n is the Fourier matrix. Then

$$f(t) = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} y_k e^{i2\pi kt}$$

satisfies $f(t_j) = x_j$, for $j = 0, 1, \dots, n-1$.

The coefficients y_k of the discrete Fourier transform are the coefficients of an interpolating function $f(t)$ in a trigonometric basis.

proof by the inverse DFT

$$\mathbf{y} = F_n \mathbf{x}, \quad f(t) = \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} y_k e^{i2\pi kt}, \quad f(t_j) = x_j, \quad t_j = j/n, \quad j = 0, 1, \dots, n$$

Proof: we use $\mathbf{x} = F_n^{-1} \mathbf{y}$, for $j = 0, 1, \dots, n-1$:

$$\begin{aligned} x_j &= \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} (\omega^{-k})^j y_k, \quad \omega = e^{-i2\pi/n} \\ &= \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} (e^{i2\pi k j/n}) y_k \\ &= \frac{1}{\sqrt{n}} \sum_{k=0}^{n-1} (e^{i2\pi k t_j}) y_k = f(t_j). \end{aligned}$$

Q.E.D.

the 8th roots of unity again

Exercise 5:

Verify the DFT interpolation property for $n = 8$.

- 1 Generate a random vector $\mathbf{x}$ of size 8.
- 2 Compute $\mathbf{y} = F_n \mathbf{x}$, with your `FourierMatrix` of Exercise 2.
- 3 Define the function $f(t)$.
- 4 Verify that $f(j/n) = x_j$, for $j = 0, 1, \dots, n - 1$.

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applied to filter design

Theorem (amplitude gain and phase shift of filter)

Let $H(z) = \sum_{k=0}^{\infty} h_k z^{-k}$ be the transfer function of a filter F .

For input $u = \left\{ u_k = \sin(\omega kT) \right\}_{k=0}^{\infty}$, $y = \left\{ y_k = r \sin(\omega kT + \phi) \right\}_{k=0}^{\infty}$ is the output, where ($\omega = 2\pi n$, T is the sampling rate),

- $r = |H(e^{i\omega T})|$ is the amplitude gain, and
- $\phi = \arg H(e^{i\omega T})$ is the phase shift.

Filter design in three steps:

- 1 Make the desired gain $r = r(t)$ and phase shift $\phi = \phi(t)$.
- 2 Evaluate the desired gains and phase shifts at equidistant angles $\theta_k \in [0, 2\pi]$, $r_k = r(\theta_k)$, $\phi_k = \phi(\theta_k)$, $\hat{h}_k = r_k e^{i\phi_k}$.
- 3 $h = \text{iDFT}(\hat{h})$ is the impulse response, which defines $H(z)$.

summary and bibliography

We defined the Discrete Fourier Transform (DFT), derived the DFT Convolution Theorem and proved the DFT Interpolation Theorem.

The main references for this lecture:

- Charles R. MacCluer:
Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.

We started Chapter 4.

- Timothy Sauer: *Numerical Analysis*, second edition, Pearson, 2012.

Chapter 10 deals with the discrete Fourier transform.